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USER MANUAL

LSafe FRed-IO library for redundant I/O modules for TIA Portal STEP 7

STEP 7 Basic/Professional V18 (TIA PORTAL) / V1.0 / LSafe FRed-IO

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1. Overview of the library

1.1. General

The "LSafe_FRed-IO" library contains fail-safe blocks for easy handling of fail-safe redundant IO and distributed IO stations.

The program tools consist of three function blocks (FBs). Those blocks are summarized and distributed as an unrestricted library (LSafe_FRed-IO).

Intended use

- The usage of the program blocks is intended for two redundant F-DIs, F-DQs or F-AIs
- Application up to SIL 3 according to IEC 62061, PL e, Cat. 4 according to EN ISO 13849-1
- The overall F-SW signature confirms an unchanged safety program. The assumptions made below only apply if the "Collective F-SW-Signature" remains unchanged. The "Collective-F-Signature", "Collective F-HW-Signature" and "F-Communication Address Signature" remain unaffected by this.
- The operating and function manuals as well as the product information for the corresponding F-CPU used must also be complied with.
- Use only by trained and experienced personnel.

Possibilities for bypassing the safety function

- Project change:
Changes to the code can bypass the safety function unnoticed. A modification of the code provided means that the specified SIL/PL consideration becomes invalid. An adaptation of the modules entails a reassessment of the certification.
- Incorrect configuration of the system:
Parameters and their meaning are clearly described in this manual to prevent incorrect configuration.

1.2. Hardware and software requirements

In order to be able to use the functionality of the library described here, the following hardware and software requirements must be met.

Hardware requirements

All blocks (FB, FC, DB, ...) of the library can be used without restriction with the following controllers:

- SIMATIC S7-1XXX product family (tested with S7-1500 Firmware V3.0)
- Simulation with SIMATIC S7-PLCSIM (from V18)
- Qualified sensors and actuators shall be used – selection of the correct components is left to the user

The environmental conditions are derived from the standard environmental conditions for the above hardware portfolio and can be found in the corresponding manuals.

Software requirements

- SIMATIC STEP 7 Basic/Professional (TIA Portal V18 or higher)
- SIMATIC STEP 7 Safety Basic/Advanced V18 or higher

NOTE	With equivalent parameterization, components other than the SIMATIC ET 200SP components shown here can also be used.
NOTE	In case of using the ET 200MP, the wire break detection of the ET 200MP F-DQ shall not be activated. After passivation of an F-DQ, there is a possibility that it cannot be reintegrated due to the wire break diagnosis.
NOTE	The startup behavior of the program blocks depends on the value status of the modules. Whether an automatic restart is permitted is determined by a risk assessment. This must be carried out depending on the application for the system used, which must be pointed out to the user (manual).
NOTE	There is no acknowledgement for program blocks needed since the acknowledgement of the modules shall be used.
NOTE	Safe state: per default 0-signal represents the safe state (closed-circuit principle is assumed) – The safe state is always application dependent. This is to be checked by the user.

1.3. Evaluation of redundant input signals

Overview

There are two types of "Sensor evaluation":

- 1oo1 evaluation
- 1oo2 evaluation

1oo1 evaluation

With the 1oo1 evaluation, there is one sensor, and it is connected to the F-module via a single channel.

1oo2 evaluation, equivalent/non-equivalent

With a 1oo2 evaluation, two input channels are occupied by:

- One two-channel sensor
- Two single-channel sensors

The input signals are compared internally.

In the case of a digital input module, two types of 1oo2 evaluation can take place: equivalent or non-equivalent.

Discrepancy analysis

When using a two-channel sensor or two one-channel sensors which measure the same process variable, the sensors interact with a slight time delay due to the limited precision of their arrangement.

The discrepancy analysis for equivalence/non equivalence is used for fail-safe applications to prevent errors from time differences between two signals for the same function. The discrepancy analysis is initiated when different levels are detected in two associated input signals (when testing for non equivalence: the same levels). A check is made to determine whether the difference in levels (when testing for non equivalence: the same levels) has disappeared after an assignable time period, the so-called discrepancy time. If not, this means that a discrepancy error exists.

For all two-channel applications the discrepancy analysis is done directly by the F-modules.

2. Working with the library

2.1. General

The blocks are stored as a type in the library. This allows you to benefit from the following advantages:

- Central update function of library items
- Versioning of library elements

NOTE

Information on general library handling can be found in the library handling guide <https://support.industry.siemens.com/cs/ww/en/view/109747503> and in the programming guide for S7-1200/1500 in the “Libraries” chapter <https://support.industry.siemens.com/cs/ww/en/view/81318674>

More information on libraries in the TIA Portal:

- Libraries topic page <https://support.industry.siemens.com/cs/ww/en/view/109738702>
- How do you open libraries in STEP 7 (TIA Portal)? <https://support.industry.siemens.com/cs/ww/en/view/37364723>
- Automate in less than 10 minutes TIA Portal: Time Savers - Global Libraries <https://support.industry.siemens.com/cs/ww/en/view/78529894>
- Which elements from STEP 7 (TIA Portal) can be stored in a library as a type or as a master copy? <https://support.industry.siemens.com/cs/ww/en/view/109476862>
- How can you automatically open a global library when starting TIA Portal V13 and higher and use it as a company library, for example? <https://support.industry.siemens.com/cs/ww/en/view/100451450>
- Library with PLC data types (LPD) for STEP 7 (TIA Portal) and SIMATIC S7-1200 / S7-1500 <https://support.industry.siemens.com/cs/ww/en/view/109482396>

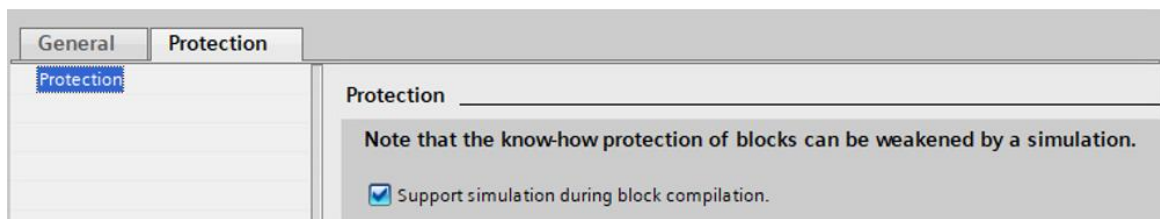
2.2. Simulatability with SIMATIC S7-PLCSIM Advanced

The simulation with SIMATIC S7-PLCSIM Advanced is deactivated in the properties of the library.

To ensure that the function blocks can be simulated after compilation with SIMATIC S7-PLCSIM Advanced, proceed as follows:

1. Open the properties of your project (Select the project in the project navigation, open the context menu by clicking on the right mouse button [alternatively the second mouse button] and select “Properties...”)
2. In the “Protection” tab, activate the option “Support simulation during block compilation”.

Figure 2-1: Support simulation during compilation with SIMATIC S7-PLCSIM Advanced



NOTE

Blocks with activated simulation capability occupy more storage space in the PLC.

3. Structure and design of the library

The following chapters describe all building blocks of the library "LSafe_FRed-IO".

The next image showcases how the library is structured.

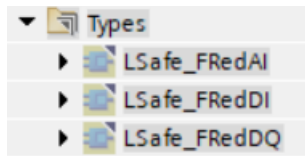


Table: Overview of signatures of the function blocks

Name	Signature	Version
LSafe_FRedDI	0x198CDC6C	V1.0
LSafe_FRedDQ	0xCD8EAA9	V1.0
LSafe_FRedAI	0x10C85743	V1.0

NOTE

The function of the mentioned devices is tested according to SIL 3 (IEC 61508, IEC 62061 and IEC 61511) or PLe (ISO 13849-1). The analysis applies only to the signatures mentioned in the table: "Overview: signatures of the function blocks". If the function modules are changed, the assumptions of the safety assessment cannot be fulfilled, because a correct function can no longer be ensured. A change can be detected by the signature of the blocks.

4. Program/Function blocks

The following chapters describe all building blocks of the library "LSafe_FRed-IO".

4.1. LSafe_FRedDI

4.1.1. Function Description

The block receives and compares two redundantly configured F-DI values:

- If both digital inputs have a valid value state (value state = 1), the output "value" is set with the process value.
- If just one of the digital inputs has a valid value state (value state = 1), the output "value" is set with the process value of the valid channel/module. A warning is set.
- If both digital inputs have an invalid value state (value state = 0), the output "value" is set to a 0-signal, and an error is set.

4.1.2. Triggering Criteria and Thresholds

No.	Triggering criterion	Component reaction
1.	One module shows invalid value state (one value state = 0).	value := process value of valid channel error := 0 warning := 1
2.	Both modules show invalid values states (both value states = 0).	value := 0 error := 1 warning := 1
3.	Both values states show valid values (both value states = 1), but one signal is true and the other one is false.	value := 1 error := 0 warning := 1

Acknowledgement and restart after faults

Whether an automatic restart is permitted is determined by a risk assessment. This must be carried out depending on the application for the system used.

The program blocks behave as followed:

- No Acknowledgement for program blocks needed since the acknowledgement of the modules shall be used
- Startup behavior of the program blocks is depending on the value status of the modules

4.1.3. Block parameters

Block interface

LSafe_FRedDI (FB)			
Bool	channel1	value	Bool
Bool	channel2	valid	Bool
Bool	valueStatusCh1	warning	Bool
Bool	valueStatusCh2	error	Bool
Time	discrepancyTime	diag	Word

Input parameters

Identifier	Data type	Default value	Description
channel1	Bool	false	Input signal of the 1 st redundant module.
channel2	Bool	false	Input signal of the 2 nd redundant module.
valueStateCh1	Bool	false	Value state of the input signal of the 1 st redundant module for instructions on diagnostics.
valueStateCh2	Bool	false	Value state of the input signal of the 2 nd redundant module for instructions on diagnostics.
discrepancyTime	Time	T#20ms	Time in which the inputs of <i>channel 1</i> and <i>channel 2</i> can differ from each other before a discrepancy error is triggered (20ms ... 30s).

Output parameters

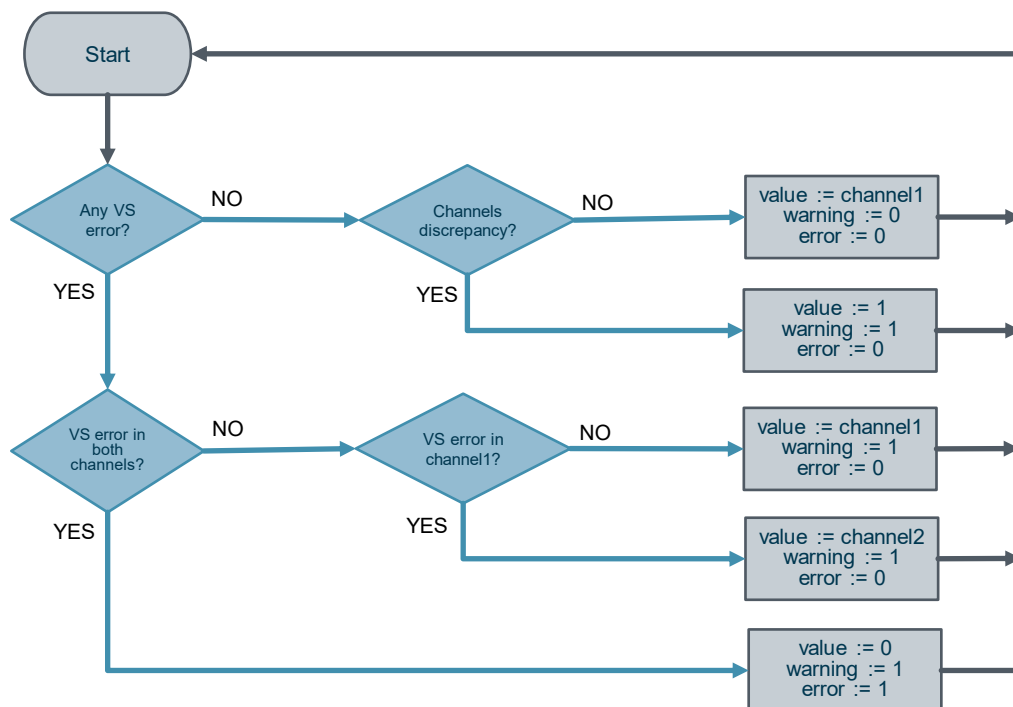
Identifier	Data type	Descriptions
value	Bool	Output of the valid channel.
valid	Bool	TRUE: The output value is valid.
warning	Bool	TRUE: Warning if a channel has become invalid.
error	Bool	TRUE: Error if both channels are invalid.
diag	Word	Diagnostic information of the block.

Diagnostic information (WORD)

Bit	Definition	Description
0.	valueStatusCh1 valid	Value state of channel 1 (1 st redundant module) is 1.
1.	valueStatusCh2 valid	Value state of channel 2 (2 nd redundant module) is 1.
2.	reserved	-
3.	discrepancy error between <i>channel 1</i> and <i>channel 2</i>	When both channels/modules are valid and their input values differ longer than the set time.
4.	reserved	-
5.	reserved	-
6.	reserved	-
7.	reserved	-
8.	reserved	-
9.	reserved	-
10.	reserved	-

Bit	Definition	Description
11.	reserved	-
12.	redundancy OK	No value state errors. No discrepancy errors.
13.	reserved	-
14.	error	There is a value state error in both modules.
15.	warning	Value state error in only one module.

4.1.4. Functional diagram



4.1.5. Maximum block runtime

CPU-Type	Runtime [ms]
121x FC	1.6
1510 SP F-1 PN	0.5
1511 F-1 PN / 1511 TF-1 PN	0.5
1515 F-2 PN / 1515 TF-2 PN	0.2
1518 F-4 PN/DP (T, ODK bzw. MEP)	0.1

4.2. LSafe_FRedDQ

4.2.1. Function description

The block receives a process value that must be written in two F-DQ modules:

- If both DQ-modules have a valid value state (value state = 1), the process value is written to both digital outputs.
- If just one of the DQ-modules has a valid value state (value state = 1), the process value is attempted to be written to both digital outputs. However, only the module with the valid value state will successfully output the value. A warning is set.
- If both DQ-modules have an invalid value state (value state = 0), both digital outputs are set to a 0-signal, and an error is set.

4.2.2. Triggering Criteria and Thresholds

No.	Triggering criterion	Component reaction
1.	One module shows invalid value state (one value state = 0).	channel1 := process value channel2 := process value error := 0 warning := 1
2.	Both modules show invalid values states (both value states = 0).	channel1 := 0 channel2 := 0 error := 1 warning := 1

Acknowledgement and restart after faults

Whether an automatic restart is permitted is determined by a risk assessment. This must be carried out depending on the application for the system used.

The program blocks behave as follows:

- No Acknowledgement for program blocks needed since the acknowledgement of the modules shall be used.
- Startup behavior of the program blocks depends on the value status of the modules.

4.2.3. Block parameters

Block interface

LSafe_FRedDQ (FB)			
Bool	value	channel1	Bool
Bool	valueStatusCh1	channel2	Bool
Bool	valueStatusCh2	valid	Bool
		warning	Bool
		error	Bool
		diag	Word

Input parameters

Identifier	Data type	Default value	Description
value	Bool	false	Signal to be output redundantly.
valueStatusCh1	Bool	false	Value state of the output signal of the 1 st redundant module for instructions on diagnostics.
valueStatusCh2	Bool	false	Value state of the output signal of the 2 nd redundant module for instructions on diagnostics.

Output parameters

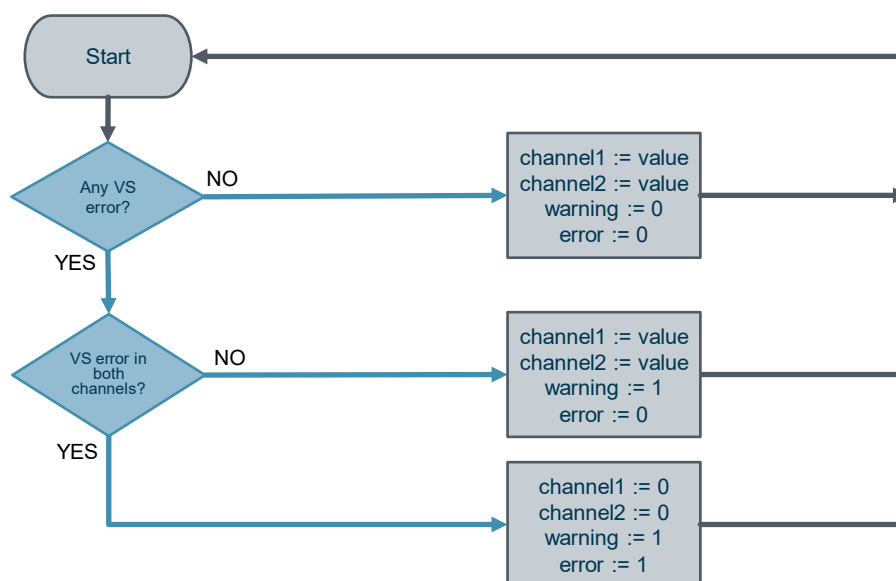
Identifier	Data type	Descriptions
channel1	Bool	Signal output for the 1 st redundant module.
channel2	Bool	Signal output for the 2 nd redundant module.
valid	Bool	TRUE: The output value is valid.
warning	Bool	TRUE: Warning if a channel has become invalid.
error	Bool	TRUE: Error if both channels are invalid.
diag	Word	Diagnostic information of the block.

Diagnostic information (WORD)

Bit	Definition	Description
0.	valueStatusCh1 valid	Value state of channel 1 (1 st redundant module) is 1.
1.	valueStatusCh2 valid	Value state of channel 2 (2 nd redundant module) is 1.
2.	reserved	-
3.	reserved	-
4.	reserved	-
5.	reserved	-
6.	reserved	-
7.	reserved	-
8.	reserved	-
9.	reserved	-
10.	reserved	-
11.	reserved	-
12.	redundancy OK	No value state errors. No discrepancy errors.

Bit	Definition	Description
13.	reserved	-
14.	error	There is a value state error in both modules.
15.	warning	Value state error in only one module.

4.2.4. Functional diagram



4.2.5. Maximum block runtime

CPU-Type	Runtime [ms]
121x FC	1.2
1510 SP F-1 PN	0.4
1511 F-1 PN / 1511 TF-1 PN	0.4
1515 F-2 PN / 1515 TF-2 PN	0.2
1518 F-4 PN/DP (T, ODK bzw. MEP)	0.1

4.3. LSafe_FRedAI

4.3.1. Function Description

The block receives and compares two redundantly configured F-AI values:

- If both analog inputs have a valid value state (value state = 1), the larger value is used as the output “value”.

- If just one of the analog inputs has a valid value state (value state = 1), the output “value” is set with the process value of the valid channel/module. A warning is set.
- If both analog inputs have an invalid value state (value state = 0), the output “value” is set to a 0-signal, and an error is set.

4.3.2. Triggering Criteria and Thresholds

No.	Triggering criterion	Component reaction
1.	One module shows invalid value state (one value state = 0).	value := process value of valid channel error := 0 warning := 1
2.	Both modules show invalid values states (both value states = 0).	value := substitutionValue, if substitutionSelection is 0. value := lastValidValue, if substitutionSelection is 1. error := 1 warning := 1
3.	Both values are valid, but they exceed the discrepancy limit.	value := maximum value between channel1 and channel2 error := 0 warning := 1
4.	Any parametrization error	value := 0 error := 1 warning := 1

Acknowledgement and restart after faults

Whether an automatic restart is permitted is determined by a risk assessment. This must be carried out depending on the application for the system used.

The program blocks behave as followed:

- No Acknowledgement for program blocks needed since the acknowledgement of the modules shall be used
- Startup behavior of the program blocks is depending on the value status of the modules

4.3.3. Block parameters

Block interface

LSafe_FRedAI (FB)			
Int	channel1	value	Int
Int	channel2	valid	Bool
Bool	valueStatusCh1	warning	Bool
Bool	valueStatusCh2	error	Bool
Int	hysteresis	diag	Word
Int	discrepancy		
Time	discrepancyTime		
Int	beginOfNominalRange		
Int	substitutionValue		
Bool	substitutionSelection		

Input parameters

Identifier	Data type	Default value	Description
channel1	Int	0	Input signal of the 1 st redundant module.
channel2	Int	0	Input signal of the 2 nd redundant module.
valueStatusCh1	Bool	false	Value state of the input signal of the 1 st redundant module for instructions on diagnostics.
valueStatusCh2	Bool	false	Value state of the input signal of the 2 nd redundant module for instructions on diagnostics.
hysteresis	Int	20	This variable defines the threshold, expressed as a percentage (0% - <i>discrepancy</i>-%), for reintegrating the system after a discrepancy error has occurred. A discrepancy error is considered resolved and the system returns to normal operation when: $\frac{ \text{redAI1} - \text{redAI2} }{27648} < \frac{\text{hysteresis}}{100}$ Explanation: If the percentage difference between the input signal of the redundant AI modules is less than the hysteresis in percent, it is enabled again.
discrepancy	Int	20	This variable defines the maximum allowed difference, expressed as a percentage (0% - 20%), between the inputs before a discrepancy error is triggered. A discrepancy error occurs when: $\frac{ \text{redAI1} - \text{redAI2} }{27648} > \frac{\text{discrepancy}}{100}$ Explanation: If the percentage difference between the input signal of the redundant AI modules is greater than the discrepancy in percent, an error is triggered.
discrepancyTime	Time	T#20ms	Time in which the inputs of <i>channel 1</i> and <i>channel 2</i> can differ from each other before a discrepancy error is triggered (20ms ... 30s).
beginOfNominalRange	Int	0	Determines the lower limit of the nominal range. Check in the F-module manual: https://support.industry.siemens.com/cs/ww/en/view/109806400
substitutionValue	Int	0	Replacement value in case of error in both channels (0-27648).
substitutionSelection	Bool	false	Use replacement value: false = substitutionValue true = Last valid value.

Output parameters

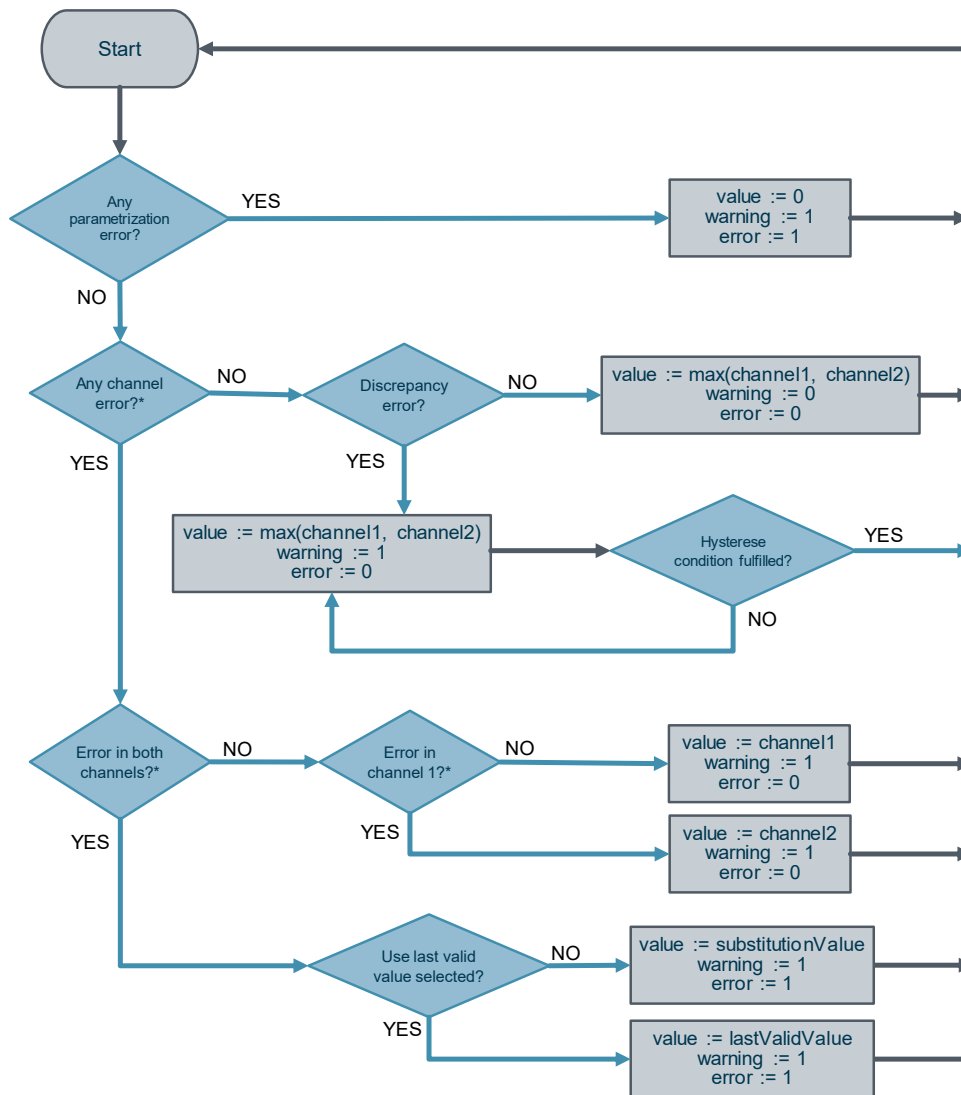
Identifier	Data type	Descriptions
value	Int	Output of the valid channel.
valid	Bool	TRUE: The output value is valid.
warning	Bool	TRUE: Warning if a channel has become invalid.
error	Bool	TRUE: Error if both channels are invalid.

Identifier	Data type	Descriptions
diag	Word	Diagnostic information of the block.

Diagnostic information (WORD)

Bit	Definition	Description
0.	valueStatusCh1 valid	Value state of channel 1 (1 st redundant module) is 1.
1.	valueStatusCh2 valid	Value state of channel 2 (2 nd redundant module) is 1.
2.	reserved	-
3.	discrepancy error between <i>channel1</i> and <i>channel 2</i>	When both channels/modules are valid and their values differ by more than the allowed discrepancy range.
4.	reserved	-
5.	reserved	-
6.	input value overrange	The value of a channel is over its nominal range (> 27648).
7.	input value underrange	The value of a channel is under its nominal range (< beginOfNominalRange).
8.	reserved	-
9.	parameter error	Any of the following values is not within its parametrization limits: · discrepancy [0, 20] · hysteresis [0, discrepancy] · discrepancyTime [20ms, 30s] · substitutionValue [0, 27648]
10.	substitute value active	The system displays the substitute value due to an error in both channels.
11.	last valid value active	The system displays the last valid value due to an error in both channels.
12.	redundancy OK	No value state errors. No over-under nominal range errors. No discrepancy errors. No parametrization errors.
13.	reserved	-
14.	error	There is a value state error in both modules. A module error could be caused by a value state error or an over-under nominal range error.
15.	warning	There is an error in only one module. A module error could be caused by a value state error or an over-under nominal range error.

4.3.4. Functional diagram



* A channel error can occur due to:
 - A value state error
 - The channel value being outside of the nominal range

4.3.5. Maximum block runtime

CPU-Type	Runtime [ms]
121x FC	2.3
1510 SP F-1 PN	0.6
1511 F-1 PN / 1511 TF-1 PN	0.6
1515 F-2 PN / 1515 TF-2 PN	0.2
1518 F-4 PN/DP (T, ODK bzw. MEP)	0.1

5. Recommendations for possible wiring variants

Within this chapter, typical wiring examples and the maximum achievable SIL / PL are shown.
The base is the manual of the F-I/O ET 200SP modules. All the warnings and notes of the manual also apply in this document.

Hyperlink to F-I/O ET 200SP modules (in Device-specific information => ET 200SP fail-safe modules):

<https://support.industry.siemens.com/cs/ww/en/view/109806400>

Special aspects for redundant operation are noted in the following chapters.

NOTE

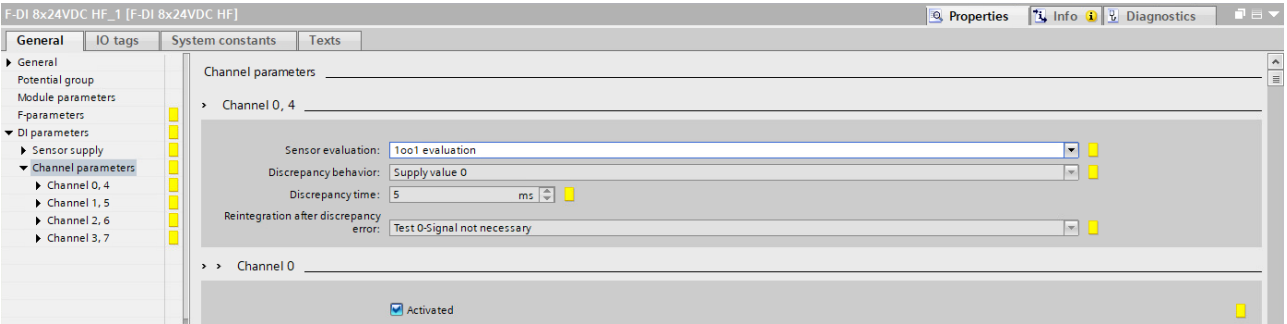
For all configurations qualified actuators and sensors shall be used.

5.1. F-DI

5.1.1. Single channel

	Non redundant sensor	Redundant sensor
Schematic structure		
Reachable SIL/Cat/PL	SIL3/Cat.3/PLd	SIL3/Cat.3/PLd
Parameters and wiring	See ET 200SP failsafe modules - F-DI module - "Application of the F-I/O module" chapter 5.1.1 in device manual. Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.	See ET 200SP failsafe modules - F-DI module - "Application of the F-I/O module" chapter 5.1.1 in device manual. Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.

To set a 1oo1 channel evaluation make the settings under "Channel parameter" as shown in the figure below:



5.1.2. Two channel

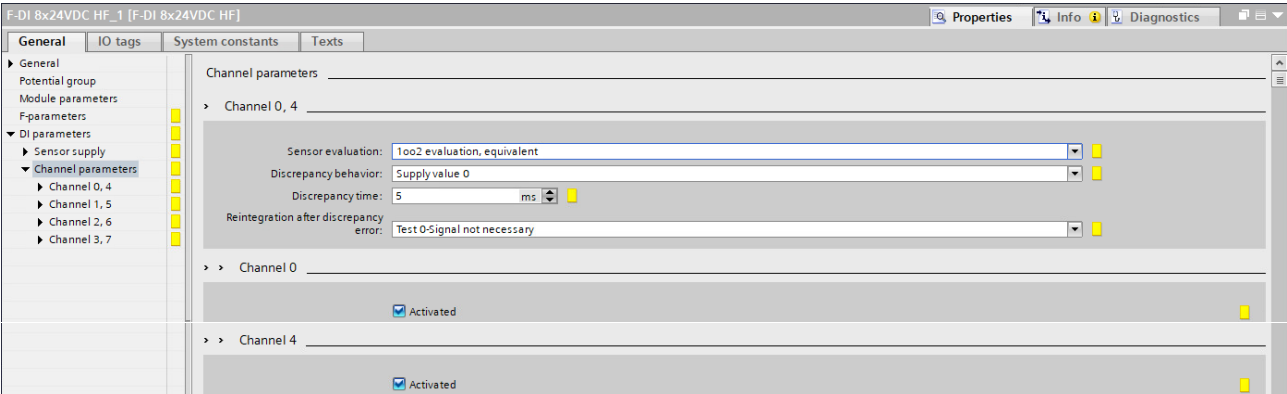
If the short-circuit test is disabled, the wiring needs to be short-circuit and cross-circuit proof or it needs to be a connection type (discrepancy, non-equivalent) which also detect the cross-circuits using discrepancy.

5.1.2.1. 1oo2 equivalent

	Non redundant sensor	Redundant sensor
Schematic structure		
Reachable SIL/Cat/PL	SIL3/Cat.4/PLe	SIL3/Cat.4/PLe
Parameters and wiring	See ET 200SP failsafe modules - F-DI module - "Application of the F-I/O module" chapter 5.1.2 in device manual Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.	See ET 200SP failsafe modules - F-DI module - "Application of the F-I/O module" chapter 5.1.3 & 5.1.3.1 in device manual Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.

Special aspects of the wiring The short circuit test must be deactivated.
The mass of both stations needs to be connected.

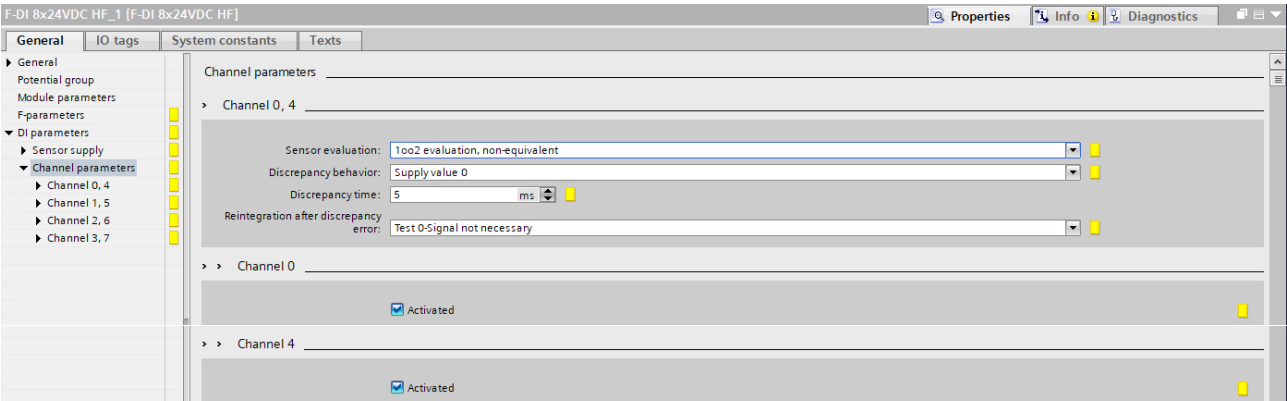
To set a 1oo2 equivalent channel evaluation make the settings under "Channel parameter" as shown in the figure below:



5.1.2.2. 1oo2 non-equivalent

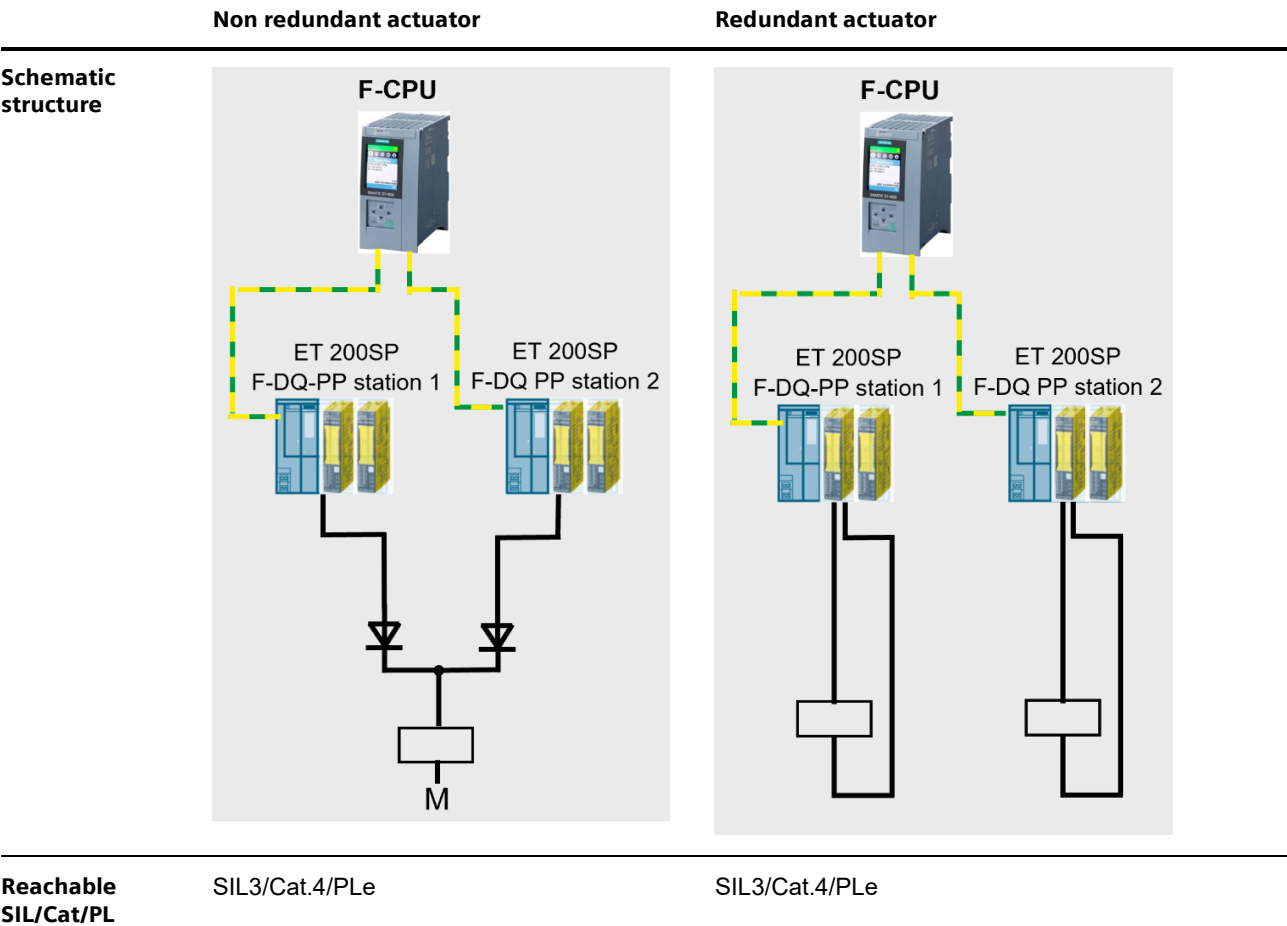
	Non redundant sensor	Redundant sensor
Schematic structure		
Reachable SIL/Cat/PL	SIL3/Cat.4/PLe	SIL3/Cat.4/PLe
Parameters and wiring	See ET 200SP failsafe modules - F-DI module - "Application of the F-I/O module" chapter 5.1.3 & 5.1.3.2 in device manual Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.	See ET 200SP failsafe modules - F-DI module - "Application of the F-I/O module" chapter 5.1.3 & 5.1.3.2 in device manual Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.
Special aspects of the wiring	The short circuit test must be deactivated. The mass of both stations needs to be connected. Short-circuits can be detected by non-equivalent interconnection type.	--

To set a 1oo2 non-equivalent channel evaluation make the settings under "Channel parameter" as shown in the figure below:



5.2. F-DQ PP

5.2.1. Single actuator



	Non redundant actuator	Redundant actuator
Parameters and wiring	See ET 200SP failsafe modules - F-DQ PP module - “Application of the F-I/O module” chapter 5.1 & 5.2 in device manual Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application. A more detailed wiring example is described in the section 6.1	See ET 200SP failsafe modules - F-DQ PP module - “Application of the F-I/O module” chapter 5.1 & 5.2 in device manual Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.

5.2.2. Two actuator

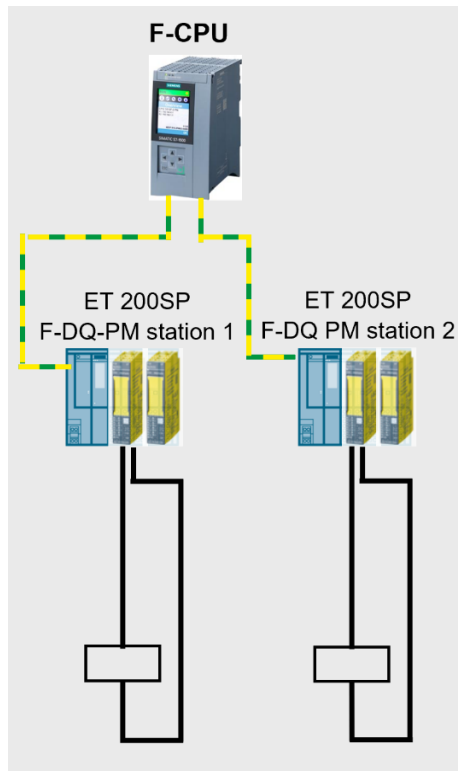
	Non redundant actuator	Redundant actuator
Schematic structure		
Reachable SIL/Cat/PL	SIL3/Cat.4/PLe	SIL3/Cat.4/PLe
Parameters and wiring	See ET 200SP failsafe modules - F-DQ PP module - “Application of the F-I/O module” chapter 5.1 & 5.3 & 5.4 in device manual Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.	See ET 200SP failsafe modules - F-DQ PP module - “Application of the F-I/O module” chapter 5.1 & 5.3 & 5.4 in device manual Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.

5.3. F-DQ PM

5.3.1. Single actuator

Redundant actuator

Schematic structure



Reachable SIL/Cat/PL

SIL3/Cat.4/PLe

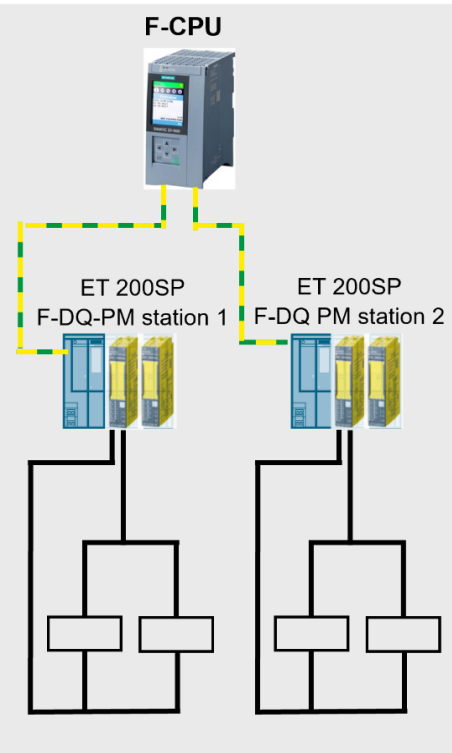
Parameters and wiring

See ET 200SP failsafe modules - F-DQ PM module - "Application of the F-I/O module" chapter 6.1 & 6.2 in device manual
Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.

5.3.2. Two actuator

Redundant actuator

Schematic structure



Reachable SIL/Cat/PL

SIL3/Cat.4/PLe

Parameters and wiring

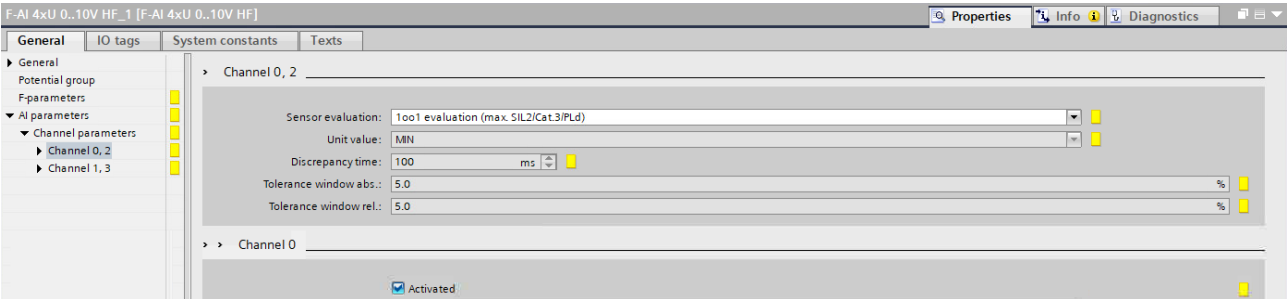
See ET 200SP failsafe modules - F-DQ PM module - “Application of the F-I/O module” chapter 6.1 & 6.3 & 6.4 in device manual
Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.

5.4. F-AI voltage measurement

5.4.1. Single channel

	Non redundant sensor	Redundant sensor
Schematic structure		
Reachable SIL/Cat/PL	SIL2/Cat.3/PLd	SIL2/Cat.3/PLd
Parameters and wiring	See ET 200SP failsafe modules - F-AI 0-10V module - "Application of the F-I/O module" chapter 5.1 & 5.3 in device manual Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application. A more detailed wiring example is described in the section 6.2	See ET 200SP failsafe modules - F-AI 0-10V module - "Application of the F-I/O module" chapter 5.1 & 5.3 in device manual Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.

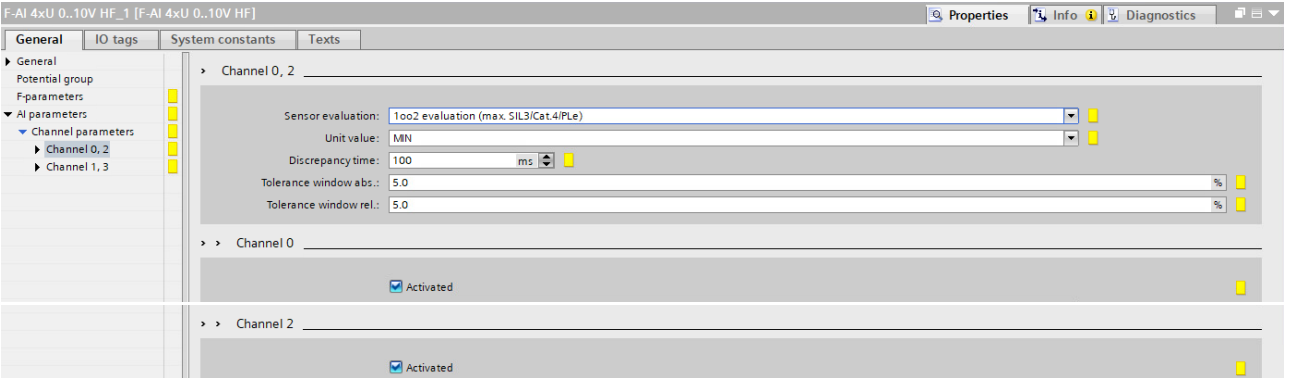
To set a 1oo1 channel evaluation make the settings under "Channel parameters" as shown in the figure below:



5.4.2. Two channel

	Non redundant sensor	Redundant sensor
Schematic structure		
Reachable SIL/Cat/PL	SIL3/Cat.4/PLe	SIL3/Cat.4/PLe
Parameters and wiring	See ET 200SP failsafe modules - F-AI 0-10V module - "Application of the F-I/O module" chapter 5.2 & 5.4 in device manual Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.	See ET 200SP failsafe modules - F-AI 0-10V module - "Application of the F-I/O module" chapter 5.2 & 5.4 in device manual Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.

To set a 1oo2 equivalent channel evaluation make the settings under "Channel parameters" as shown in the figure below:

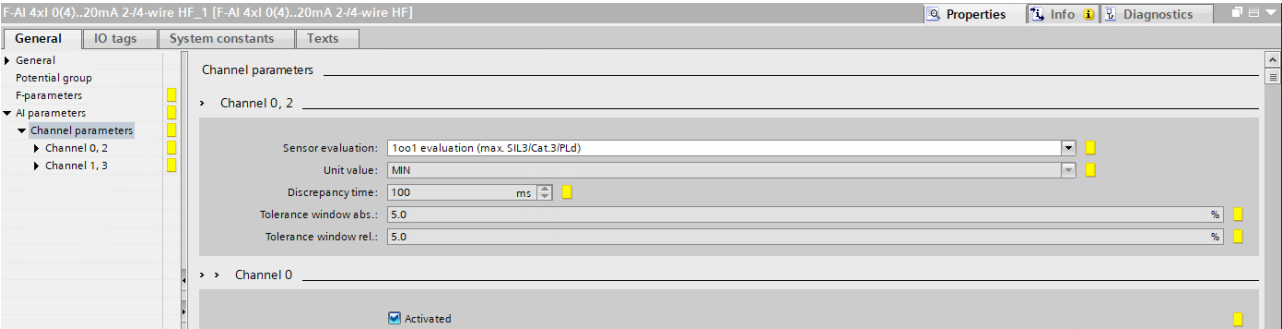


5.5. F-AI current measurement

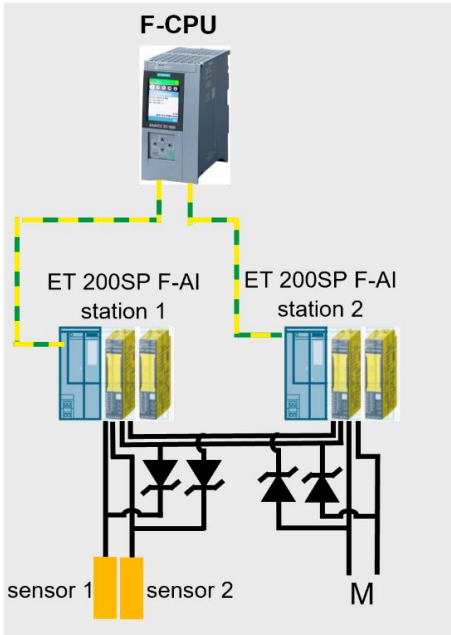
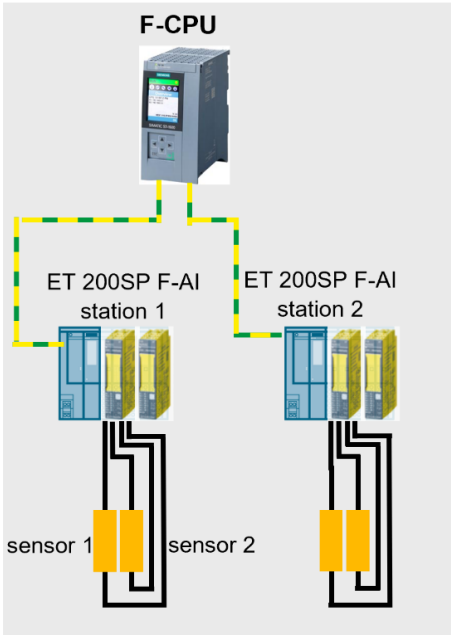
5.5.1. Single channel

	Non redundant sensor	Redundant sensor
Schematic structure		
Reachable SIL/Cat/PL	SIL3/Cat.3/PLd	SIL3/Cat.3/PLd
Parameters and wiring	See ET 200SP failsafe modules - F-AI 0(4)-20A module - "Application of the F-I/O module" chapter 5.1 & 5.3 in device manual Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application. A more detailed wiring example is described in the section 6.3	See ET 200SP failsafe modules - F-AI 0(4)-20A module - "Application of the F-I/O module" chapter 5.1 & 5.3 in device manual Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.
Note for encoder selection	Internal resistance of F-AI is 125 Ω. In redundant applications the encoder must be capable of dealing with a load of > 250 Ω.	--

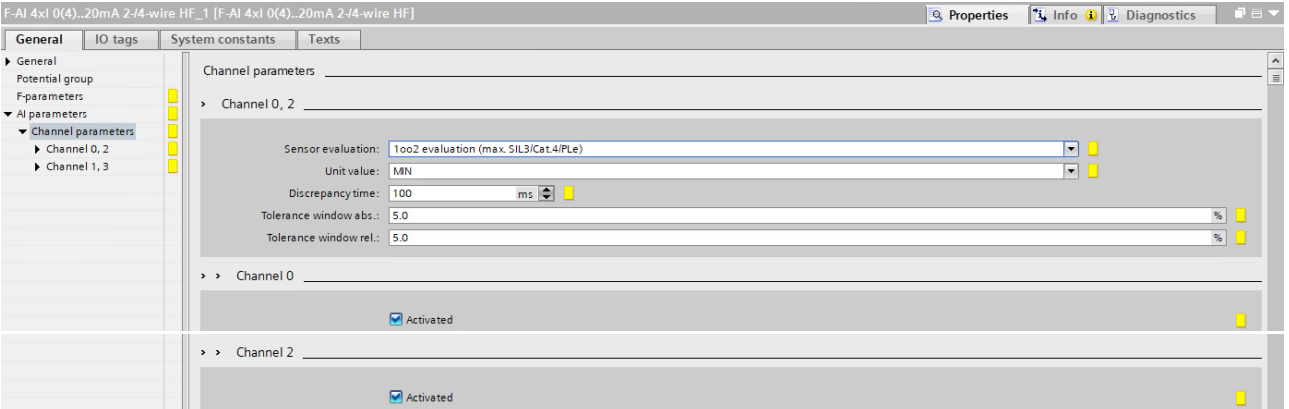
To set a 1oo1 channel evaluation make the settings under "Channel parameters" as shown in the figure below:



5.5.2. Two channel

	Non redundant sensor	Redundant sensor
Schematic structure		
Reachable SIL/Cat/PL	SIL3/Cat.4/PLe	SIL3/Cat.4/PLe
Parameters and wiring	See ET 200SP failsafe modules - F-AI 0(4)-20A module - "Application of the F-I/O module" chapter 5.2 & 5.4 in device manual Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.	See ET 200SP failsafe modules - F-AI 0(4)-20A module - "Application of the F-I/O module" chapter 5.2 & 5.4 in device manual Parameters are described there and special aspects for the wiring are mentioned. All hints and warning given in the usecase of the device manual also apply for this application.
Note for encoder selection	Internal resistance of F-AI is 125 Ω. In redundant applications the encoder must be capable to deal with a load of > 250 Ω.	--

To set a 1oo2 equivalent channel evaluation make the settings under "Channel parameters" as shown in the figure below:

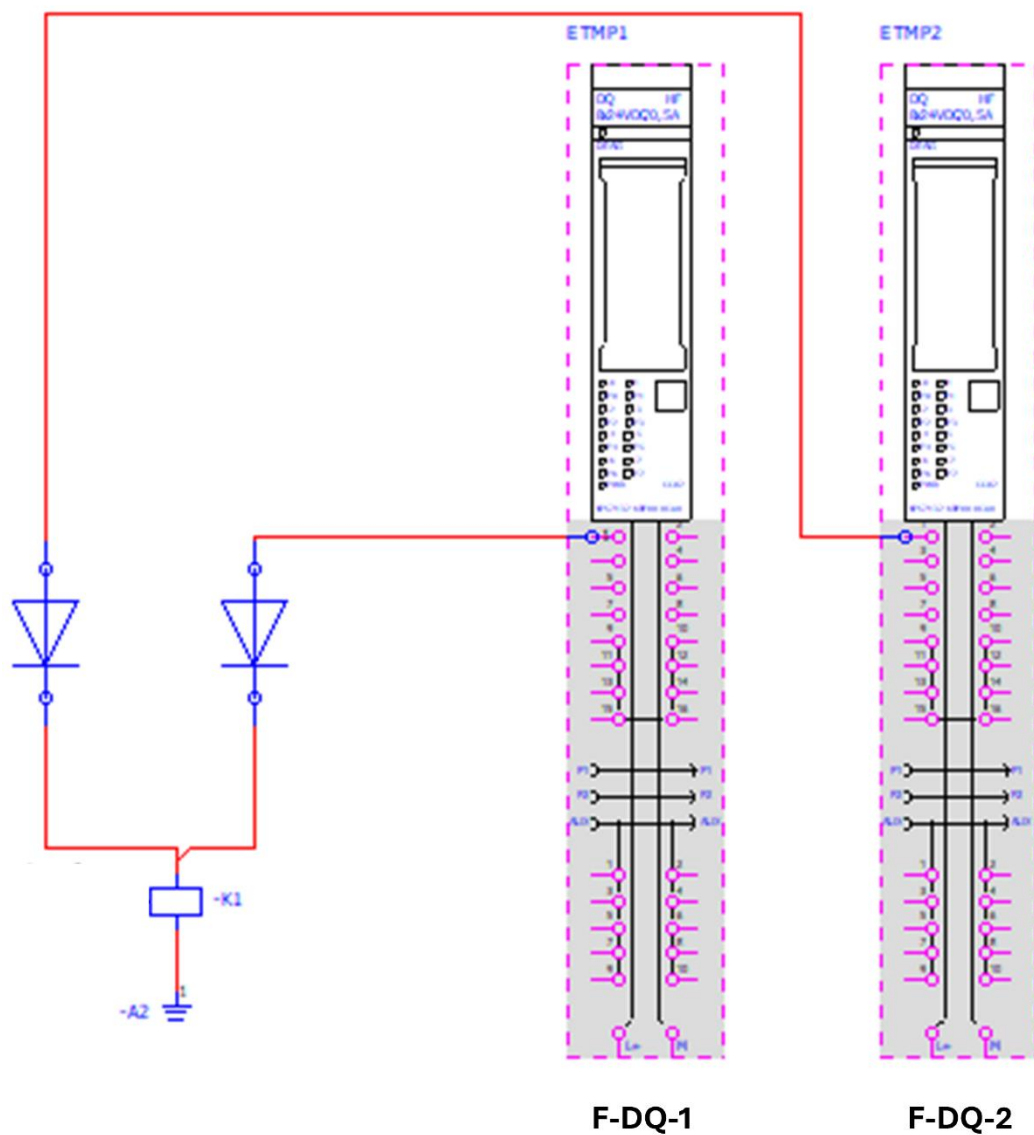


6. Further information

This library foresees the use of different wiring options to connect two F-DQ to the same actuator or to connect two F-AI channels from the same sensor to two different modules.

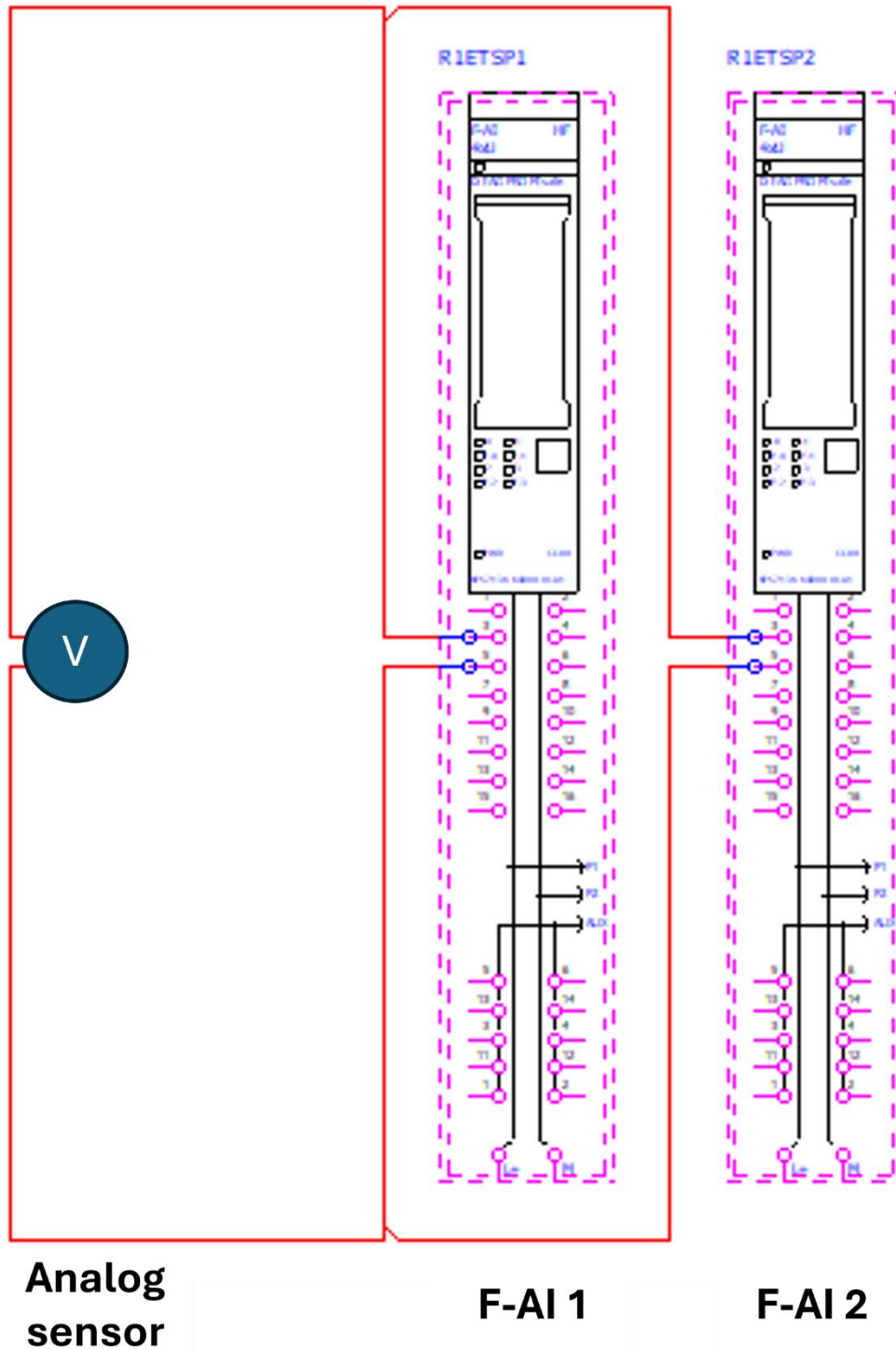
6.1. F-DQ PP single actuator

Figure 6-1: Wiring for two F-DQ channels connected to the same actuator



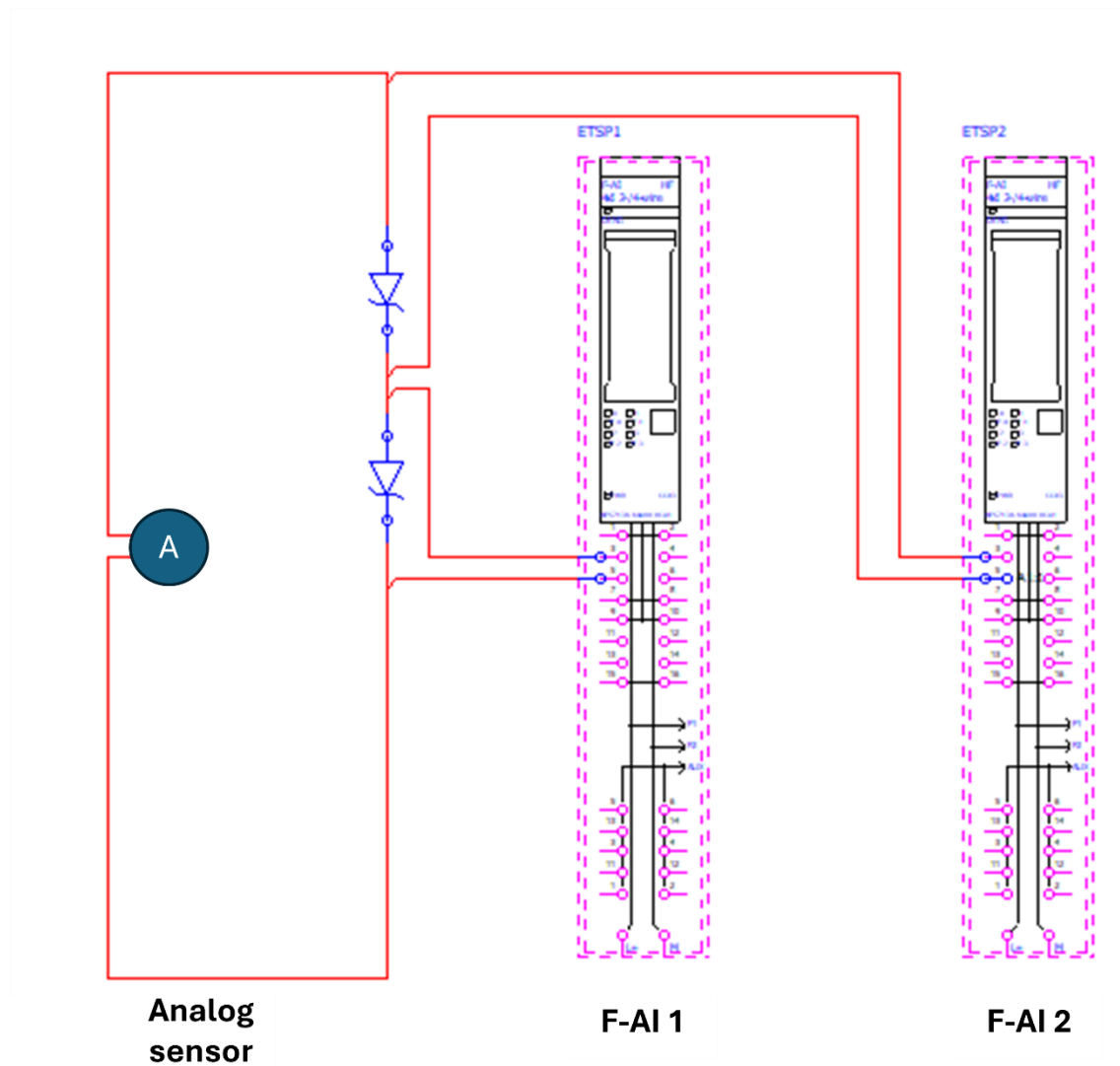
6.2. F-AI voltage single sensor

Figure 6-2: Wiring of two F-AI voltage measure channels connected to the same sensor



6.3. F-AI current single sensor

Figure 6-3: Wiring of two F-AI current measure channels connected to the same sensor



7. Appendix

7.1. Service and support

SiePortal

The integrated platform for product selection, purchasing and support - and connection of Industry Mall and Online support. The SiePortal home page replaces the previous home pages of the Industry Mall and the Online Support Portal (SIOS) and combines them.

- **Products & Services**
In Products & Services, you can find all our offerings as previously available in Mall Catalog.
- **Support**
In Support, you can find all information helpful for resolving technical issues with our products.
- **mySieportal**
mySiePortal collects all your personal data and processes, from your account to current orders, service requests and more. You can only see the full range of functions here after you have logged in.

You can access SiePortal via this address: sieportal.siemens.com

Technical Support

The Technical Support of Siemens Industry provides you fast and competent support regarding all technical queries with numerous tailor-made offers – ranging from basic support to individual support contracts.

Please send queries to Technical Support via Web form: support.industry.siemens.com/cs/my/src

SITRAIN – Digital Industry Academy

We support you with our globally available training courses for industry with practical experience, innovative learning methods and a concept that's tailored to the customer's specific needs.

For more information on our offered trainings and courses, as well as their locations and dates, refer to our web page: siemens.com/sitrain

Industry Online Support app

You will receive optimum support wherever you are with the "Industry Online Support" app. The app is available for iOS and Android:



7.2. Links and literature

No. Topic

11	Siemens Industry Online Support https://support.industry.siemens.com
12	Link to this entry page of this application example https://support.industry.siemens.com/cs/ww/en/view/109995680
13	Programming Guidelines and Programming Styleguide for SIMATIC S7-1200 and S7-1500 and WinCC (TIA Portal) https://support.industry.siemens.com/cs/ww/en/view/81318674
14	Guideline on Library Handling in TIA Portal https://support.industry.siemens.com/cs/ww/en/view/109747503
15	Libraries in the TIA Portal https://support.industry.siemens.com/cs/ww/en/view/109738702

7.3. Change documentation

Version Date Modification

V1.0	09/2025	First version
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